



## Solution Guide: Introduction to Scripts

Completing this activity required the following steps:

- Creating and moving into the directory `~/Security_scripts`.
- Writing shell script `backup.sh` to automate archives and backups.
- Writing shell script `update.sh` to automate software package updates and removal.
- **Bonus:** Writing shell script `cleanup.sh` to automate the cleanup of cached files and generate a report of system resource usage.
- Testing the scripts by running them with Bash using the `sudo ./<name of the script>.sh` command.

## Solutions

1. Begin by creating a directory to hold your scripts in `~/Security_scripts`. Then, move into this directory.
  - `mkdir -p ~/Security_scripts`
  - `cd ~/Security_scripts`
2. `backup.sh`
  - [See backup.sh for the complete script.](#)
3. `update.sh`
  - [See update.sh for the complete script.](#)
4. Make each of these custom scripts executable.
  - Run the following commands:
    - `chmod +x backup.sh`
    - `chmod +x update.sh`
5. Test the scripts by running them with Bash using the `sudo ./<name of the script>.sh` command.
  - **Note:** Since we are interacting with system directories and processes such as `apt`, we need to use `sudo` for our scripts.
  - Run the following commands:
    - `sudo ./backup.sh`
    - When testing `backup.sh`, stop the script with `Ctrl + C`. Otherwise, it will take a long time to create a full backup of `/home`. We just want to see that it successfully runs.
    - `sudo ./update.sh`

## Bonus

6. `cleanup.sh`
  - [See `cleanup.sh` for the complete script.](#)
  - Make each of these custom scripts executable.
    - Run the following commands at the command prompt as follows:
      - `chmod +x cleanup.sh`
  - Test the scripts by running them with Bash using the `sudo ./<name of the script>.sh` command.
  - Since we are interacting with system directories and processes such as `apt`, we need to use `sudo` for our scripts.
    - `sudo ./cleanup.sh`
  - **Apt cache:** After installing a package, the `apt` manager saves the package and dependencies in a cache folder. They remain after installation unless the `apt` cache is cleared.
  - **Thumbnails cache:** With each image file, your Linux distro creates a thumbnail when you view images in your file manager. Thumbnails often remain for pictures that no longer exist. Therefore, this can be beneficial to clear old thumbnails.